Kevin Chia-Ming **Chang**

PH.D. CANDIDATE | FORMAL METHODS, VERIFICATION, MACHINE LEARNING

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Summary

Ph.D. candidate at UC Berkeley building scalable formal-methods and machine-learning research for safety-critical systems. Experience spans neural network verification, contract-based design, optimization, and interpretable controller synthesis, with a focus on translating theory into solver-backed algorithms and rigorous empirical evaluation.

Skills

Research Areas	Formal verification, interpretable ML, contract-based design, reinforcement learning
Methods	SMT, MILP, assume-guarantee reasoning, symbolic abstraction, safe reachability
Programming	Python, SystemVerilog, MATLAB, C++
ML / Data	PyTorch, scikit-learn, reinforcement learning, semantic segmentation, domain adaptation
Tools	Gurobi, CVX, JasperGold, Cadence Xcelium, Genus, VCS

Research Experience

DesCyPhy Lab, Prof. Pierluigi Nuzzo — *Berkeley, CA*
Graduate Student Researcher — *01/2021 – Present*

Neural Controller Transformation for Formal Verification

— Developed a **provably exact transformation** from ReLU neural-network controllers to soft decision-tree representations, making solver-based verification more scalable and interpretable.
— Proved correctness and demonstrated closed-loop safety certification on CartPole, MountainCar, and CarRacing benchmarks; resulted in a **CDC 2023 publication**.

Robustness Contracts for Learning-Enabled CPS

— Introduced **contract-based robustness specifications** that decouple plant dynamics from neural controllers for compositional safety verification.
— Designed an **SMT-based refinement algorithm** that incrementally tightens layer-wise abstraction bounds to accelerate safe-reachability analysis under bounded uncertainty.

Assume–Guarantee Contract Optimization

— Formulated **MILP-based contract-refinement** techniques to automatically synthesize tighter assume-guarantee specifications.
— Improved realizability and reduced verification complexity in multi-agent autonomous-driving scenarios.

Deep RL Controller Distillation for Verifiability

— Developed a controller-distillation framework that extracts compact, interpretable control laws from deep RL policies.
— Applied optimization-based refinement to preserve task performance while improving closed-loop stability and formal verifiability; resulted in an **Allerton 2022 publication**.

Electronic Design Automation Lab, Prof. Yao-Wen Chang — *Taipei, Taiwan*
Undergraduate Researcher — *09/2018 – 02/2020*

Routing and Optimization for EDA Contests

— Implemented an initial detailed-routing engine with concurrent track assignment and multithreaded routing; handled designs with up to **1M nets** using less than **64 GB** of memory in 10 hours on most benchmarks.
— Built a routability-driven hybrid global router and a system-level FPGA router with TDM optimization; contributed to **2nd place at ACM ISPD 2019**, **Top 5 / Honorable Mention at ICCAD 2019**, and an **ICCAD 2021 publication**.

Education

University of California, Berkeley — *Berkeley, CA*
Ph.D. in Electrical and Computer Engineering (GPA: 3.91/4.00) — *01/2021 – Present*

- Research focus: formal verification, ML-enabled cyber-physical systems, optimization, and contract-based design
- Lab: DesCyPhy Lab, The Donald O. Pederson Center for Electronic Systems Design

National Taiwan University — *Taipei, Taiwan*
B.S. in Electrical Engineering (GPA: 3.67/4.00, Last 60: 4.00/4.00) — *09/2015 – 01/2020*

- Relevant coursework: physical design, integrated circuit design, computer architecture, machine learning, deep learning for computer vision
- Undergraduate research in EDA and formal methods

Additional Experience

Apple *Cupertino, CA*
Formal Verification Engineer Intern *05/2026 – 08/2026*
- Developed and debugged **formal properties** for CPU microarchitecture blocks, targeting control-flow correctness.
- Analyzed proof failures by refining assertions, assumptions, and coverage targets to distinguish design issues from environmental constraints.

University of California, Berkeley *Berkeley, CA*
Graduate Student Instructor of Introduction to Design Automation *01/2026 – 05/2026*
- Developed **end-to-end instructional examples** demonstrating the **EDA toolchain workflow** across synthesis, simulation, and verification.
- Supported course delivery for design automation topics spanning RTL design, logic synthesis, timing analysis, and verification workflows.

University of Southern California *Los Angeles, CA*
Graduate Student Instructor (3×) of Computer Architecture *01/2022 – 05/2023*
- Developed **instructional examples** illustrating processor pipelines and the transition from in-order to out-of-order execution.
- Supported students in reasoning about performance tradeoffs across pipelining, hazards, caching, and memory hierarchy concepts in a core computer architecture curriculum.

Institute of Information Science, Academia Sinica *Taipei, Taiwan*
Software Researcher *06/2018 – 09/2018*
- Implemented a semi-supervised domain-adaptation pipeline for semantic segmentation using unpaired image translation and segmentation modeling.
- Improved mean IoU on the Cityscapes benchmark by combining segmentation training with CycleGAN-generated domain information.

Selected Project

Sign Language Translation System *[C++/Python/PyTorch/Scikit-Learn]*
Arm Design Contest *09/2018*
- Built a real-time sign-language translation system using a single 9-DOF IMU for gesture recognition.
- Won **2nd Place** and the **Best Popularity Award** in the Arm Design Contest.

Selected Publications

1. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, "Equivalent and Compact Representations of Neural Network Controllers With Decision Trees," *IEEE Transactions on Automatic Control*, pp. 1–14, 2026.
2. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, "Exact and Cost-Effective Automated Transformation of Neural Network Controllers to Decision Tree Controllers," in *Proc. IEEE Conference on Decision and Control (CDC)*, pp. 7843–7848, 2023.
3. Nathan Dahlin, Kevin Chang, K. C. Kalagarla, Rahul Jain, and Pierluigi Nuzzo, "Practical Control Design for the Deep Learning Age: Distillation of Deep RL-Based Controllers," in *Proc. Allerton Conference on Communication, Control, and Computing*, 2022.
4. Jair Certorio, Kevin Chang, Nuno C. Martins, Pierluigi Nuzzo, and Yasser Shoukry, "Passivity Tools for Hybrid Learning Rules in Large Populations," *Automatica*, 2026.
5. Wei-Kai Liu, Ming-Hung Chen, Chia-Ming Chang, Chen-Chia Chang, and Yao-Wen Chang, "Time-Division Multiplexing Based System-Level FPGA Routing," in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2021.

Selected Awards

Outstanding Performance Scholarship, *National Taiwan University* *Taipei, Taiwan*
Awarded for outstanding academic performance in electrical engineering *11/2019*
- Recognized for strong academic performance and research contributions during undergraduate study in electrical engineering.

Top 5 and Honorable Mentions, *Cad Contest Problem C* *Westminster, USA*
LEF/DEF-based open-source global router *11/2019*
- Recognized in an ACM/IEEE contest field of 100+ teams.

2nd Place, *ACM ISPD Contest* *San Francisco, USA*
Initial detailed routing *04/2019*
- Placed 2nd among 33 international teams in a leading EDA routing competition.

2nd Place and Best Popularity Award, *Arm Design Contest* *Taipei, Taiwan*
Sign language translation system *11/2018*
- Placed in the top 3 among 130+ teams in a hardware and embedded-systems design competition.